On the behaviour of Random Walks on Regular Graphs with Applications to Randomized Depth First Search

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has been carried out by **Sivasanjai G A** under the supervision of **Dr. Sandeep Sen**, and has been approved for submission as a partial requirement for the aforementioned degree.

Date of Approval: _____

Supervisor

Dr. Sandeep Sen

Signature: _____

PAC

Department of Computer Science, and Mathematics

Signature: _____

Abstract

The question of how probabilistic processes behave on regular graphs – a class of graphs which are relatively simple, yet not explored enough to be interesting – and how those insights tie into the analysis of probabilistic search algorithms such as the randomized depth first search, is central to the work here-in presented. To this end, the unknown network exploration setting introduced by Alpern and Lidbetter is adopted, and Bloch, Dutta, and Dubinzki's proof that randomized depth first search is optimal for trees is built upon to our conjecture that rDFS, in expectation, discovers about half the nodes ($\pm\theta(\sqrt{n})$) before finding its target node on a regular graph. In search for such a proof, random walks are unlocked as a useful proxy to rDFS (whose lack of Markov property makes it notoriously difficult to analyse) with their relationship established and differences explored.

And so, the focus now turns to the behavior of random walks on torodial grids – our prefered experimental setting for regular graphs. Upon establishing basic symmetry properties, Piyush's resistence network type theoretical setting, casting our notion of cost of the walk as simply the potential difference between two nodes, which again is simply the sum of harmonic probability functions, is presented. Further exploring the matrix properties of toroids to make the computation of cost from the resistence setting more effecient led to the discovery of the block circulant property of the graph laplacian of the torous, which allows for diagonalization using 2D-Discrete Fourier Transform. This unlocks a critical insight that cost and associated probabilities behave like the lattic green's function on the grid, exhibiting a $\frac{1}{2\pi}\ln n + c + O(1)$ type asymptotic behaviour explaining the many phenomena encountered before. Finally, an involution type argument is invoked to try and understand why all this behaviour occurs around the $\frac{n-1}{2}$ mark, which is also the average expected cost across all nodes.

What started as a purely algorithmic question, eventually turned out to be an exciting excursion into the many associated areas of mathematics. And so, if nothing else, this thesis is to be understood as an excercise in deconstructing the behaviour of random walks on regular graphs through the lenses of graph theory and algebra, probability, and linear algebra and computation, with an eye for algorithmic applications.

Motivation and Background:

Searching and exploration in networks is one of the most fundamental and widely studied problems in all of computer science, tracing all the way back to work by Lucas and Tarry in the 19th century [1]. In the years since, countless different search settings, and exploration strategies have been formulated and studied. Game theoretic style formulation and analysis is a particularly popular method for analysing search strategies in a given setting. Of interest to this work is the setting and strategy introduced by Alpern and Lidbetter where in an immobile hidden object is located at an unknown point H of a known network Q; they also introduce a search paradigm known as **expanding search**. Unlike the usual paradigm (**pathwise search**) where the searcher follows a continuous unit speed path until the target is reached, under the expanding search paradigm the searcher may restart the search at any time from any previously reached point. The cost of moving between points that are already reached is negligible; the portion of the search region that has been covered by time t expands in a continuous way until the first time T that it contains the target of the search [2]. Such scenarios arise frequently in the real world; consider, for example, the scenario of mining towards coal seams where-in the cost of digging is much higher than moving the equipment within the routes that have been dug.

The Alpern-Lidbetter setting, however, assumes that the searcher knows the network before starting the search. The more interesting scenario then is the case where the searcher is tasked with searching in an unknown network and without any information about the distance to hider. This leads to the setting presented by Bloch, Dutta and, Dziubinski[3], in which, the hider, subject to a budget constraint restricting the number of links, chooses a connected network and a location of an object. Meanwhile, the seeker, not observing the network and location of the target, chooses a network exploration strategy starting at a fixed node in the network. An important point to note is that the seeker, knowing that the hider balances between having the hiding place close and hard to find, at the same time, has to take it into account when designing a good search strategy. In the same paper, they prove that **randomized Depth First Search** is optimal when the hider is only allowed enough budget to construct trees (i.e,

no cycles).

Randomized Depth-First Search

Randomized Depth-First Search Algorithm

Algorithm 1 Randomized Depth-First Search (rDFS)

Require: Graph $G = (V, E)$, Starting Node $s \in V$

Ensure: A traversal order of nodes

```
Initialize an empty stack S
Push s onto S
Initialize an empty set Visited
Initialize an empty list TraversalOrder
while S is not empty do
    v ← S.pop()
    if v ∉ Visited then
        Add v to Visited
        Append v to TraversalOrder
        neighbors ← GetNeighborsG, v
        Shuffleneighbors {Randomize the order of neighbors}
        for all u ∈ neighbors do
            if u ∉ Visited then
                Push u onto S
            end if
        end for
    end if
end while
return TraversalOrder
```

Randomized Depth-First Search (rDFS) is a variant of the traditional Depth-First Search (DFS) algorithm. While DFS explores as deeply as possible along each branch before backtracking,

rDFS introduces randomness in the order of visiting neighbouring nodes. This randomization leads to varied traversal paths across different runs, making it useful in scenarios where deterministic exploration may lead to suboptimal results. While the paper in question deals with trees and graphs with a single cycle, randomized DFS seems to have an expected cost (i.e, number of new nodes discovered before hider/target at any given location in an unknown network) of $n/2$ where n is the number of vertices in a connected regular graph, as conjectured by prof. Sandeep Sen.

Conjecture

Conjecture 1. *For $G(V, E)$, such that G is regular, for any $v \in V$,*

$$E[rDFS(v)] = \frac{n}{2} \pm \theta(\sqrt{n})$$

Over the course of experiments, this $n/2$ cost seems also to hold for various types of graphs such as trees, grids, lollipop graphs, etc. Plotting the node visitation probabilities for all nodes in a grid with respect to the search of some randomly sampled nodes seems to offer up an explanation as to why rDFS is quite effective for the given problem and how it can be optimized further. One of the stated issues in Bloch et al. is the issue with formalizing and analysing rDFS style strategies when the hider's budget is unconstrained (i.e, graphs that are highly connected/ with multiple cycles). One curious property that has been noticed during experimentation is how the cost and node visitation probabilities of rDFS and simple random walk centred at root seem to coincide. This can possibly allow for rDFS to be analysed in terms of a random walk, on whose analysis a great amount of literature exists. The same can now be presented in the following section and discussed further.

Initial Experiments and Outcomes:

Random Walk Experiment Set Up:

To understand the interplay between rDFS and random walk for searching, a few experiments were performed. In all of them, a 100 x 100 grid centred at $(50, 50)$ was considered. In order to forego boundary events, the left and right, top and bottom of the grid are simply joined into a torus. First, a heatmap corresponding to the number of new nodes discovered before hitting a given node was generated. Then, these were flattened into a histogram to understand the number of nodes falling into a rough cost (new node) bucket. These were then sorted according to distance of the node from origin and plotted. In-order to better understand the effect of distance, the visitation probability of nodes along the $L2$ and $L1$ norms were plotted. Furthermore, to get a sense of how often a given node is visited while searching for a hider node, 10 nodes were randomly sampled from the grid and random walk from origin performed till the hider node is hit; whether a given node was discovered during a random walk run was kept track of using binary values and the same was added and normalized over 10000 runs to generate probability values. This results in a probability heat map for the whole grid.

rDFS Experiment Set Up:

In line with the random walk set up, once again a 100 x 100 torus centred at $(50, 50)$ was considered. In similar fashion, cost/DFS numbering was plotted against Euclidean distance from origin. The average cost over all nodes lying at a given $L2$ distance is also plotted and considered. Visitation probabilities are calculated for randomly sampled target nodes and corresponding heatmap generated. Variation of visitation probability over distance from origin is also considered.

Discussion on Initial Outcomes:

Upon studying first the cost heat maps of both the random walk and rDFS, it is possible to notice they are quite similar in that on average less number of nodes seem to be discovered

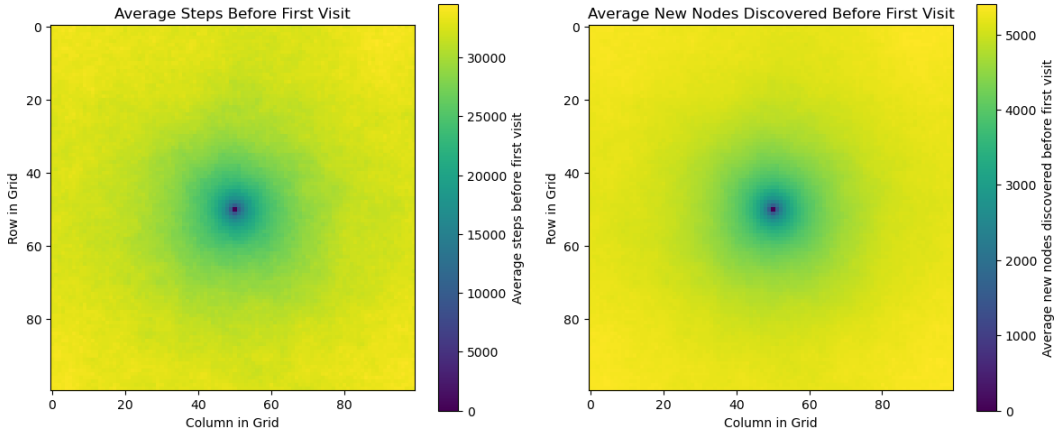

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Figure 1: Heat map of number of steps and new nodes discovered (cost) for random walk

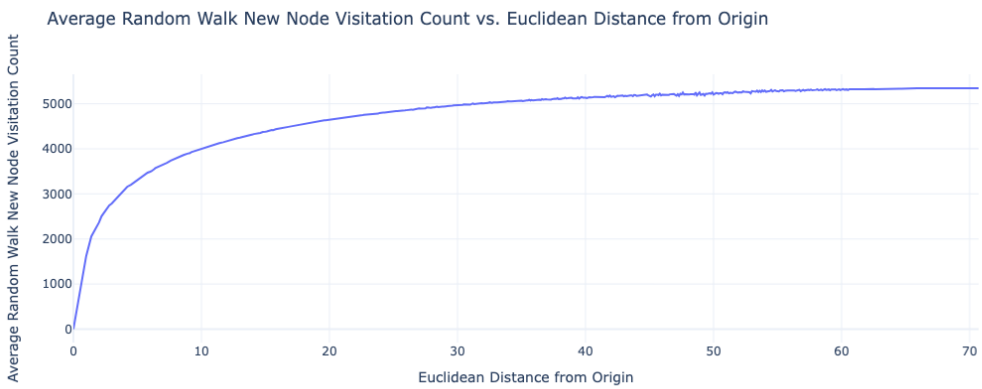

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Figure 2: Cost variation with Euclidean distance

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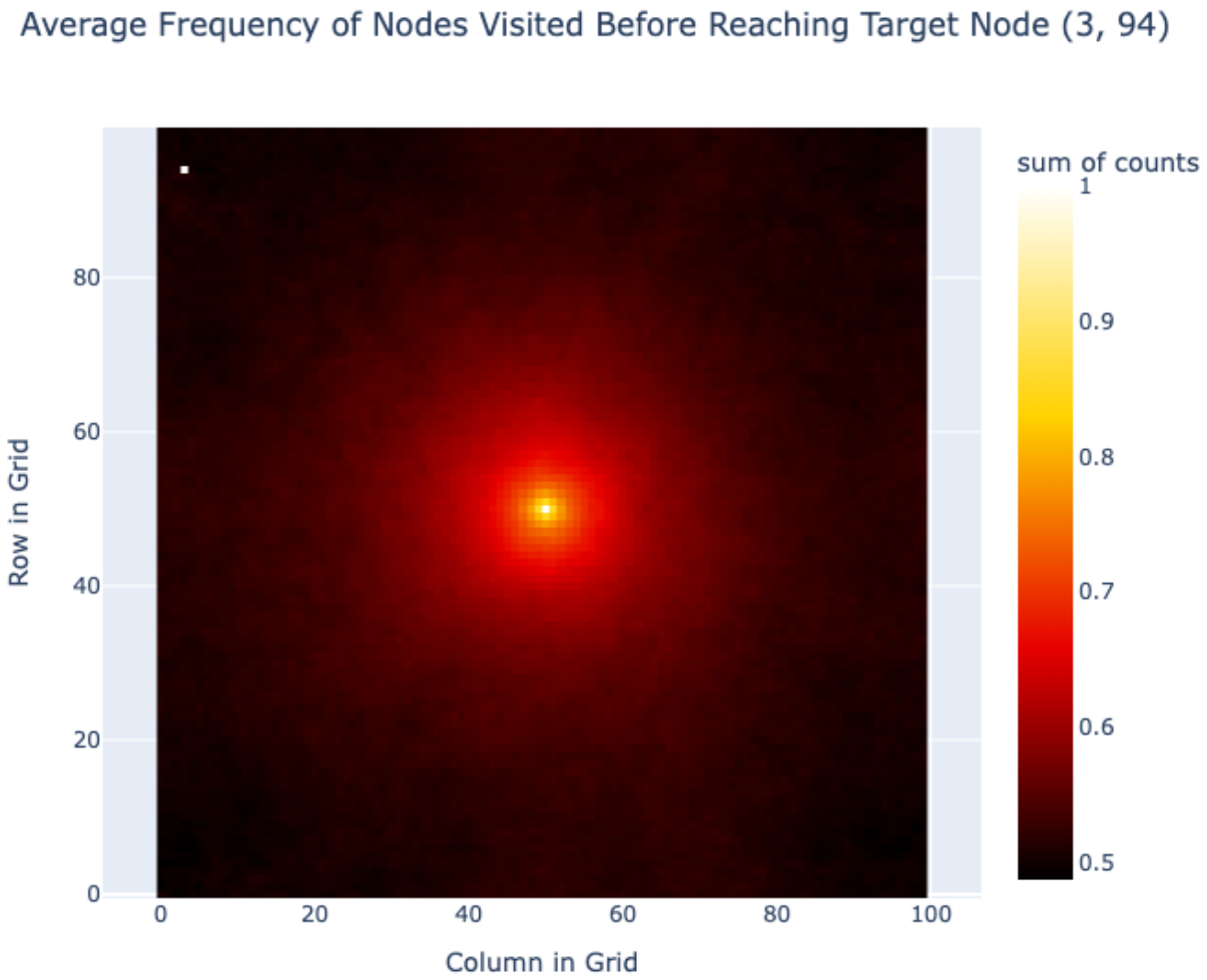

Figure 3: Visitation probabilities of nodes with respect to randomly sampled target node.

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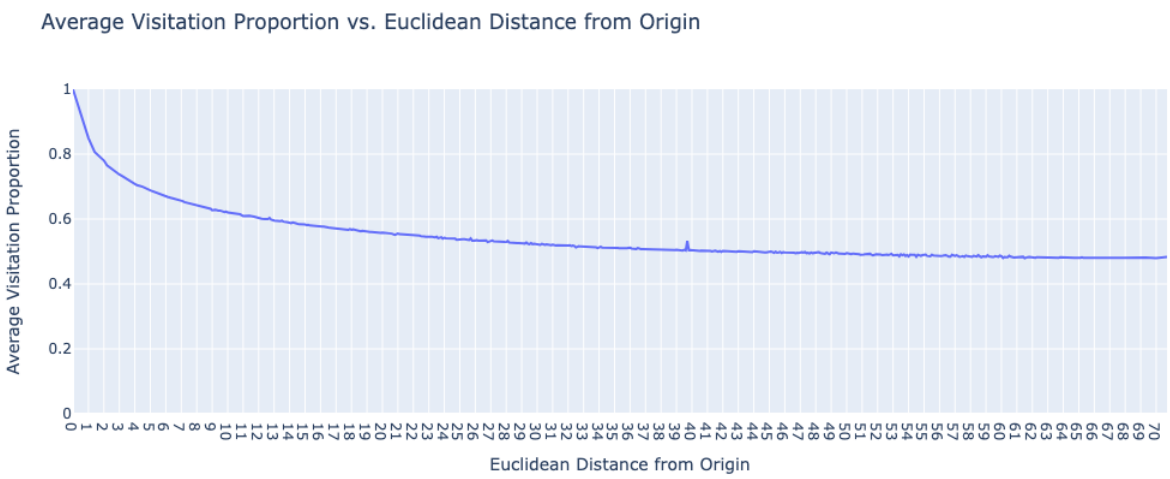


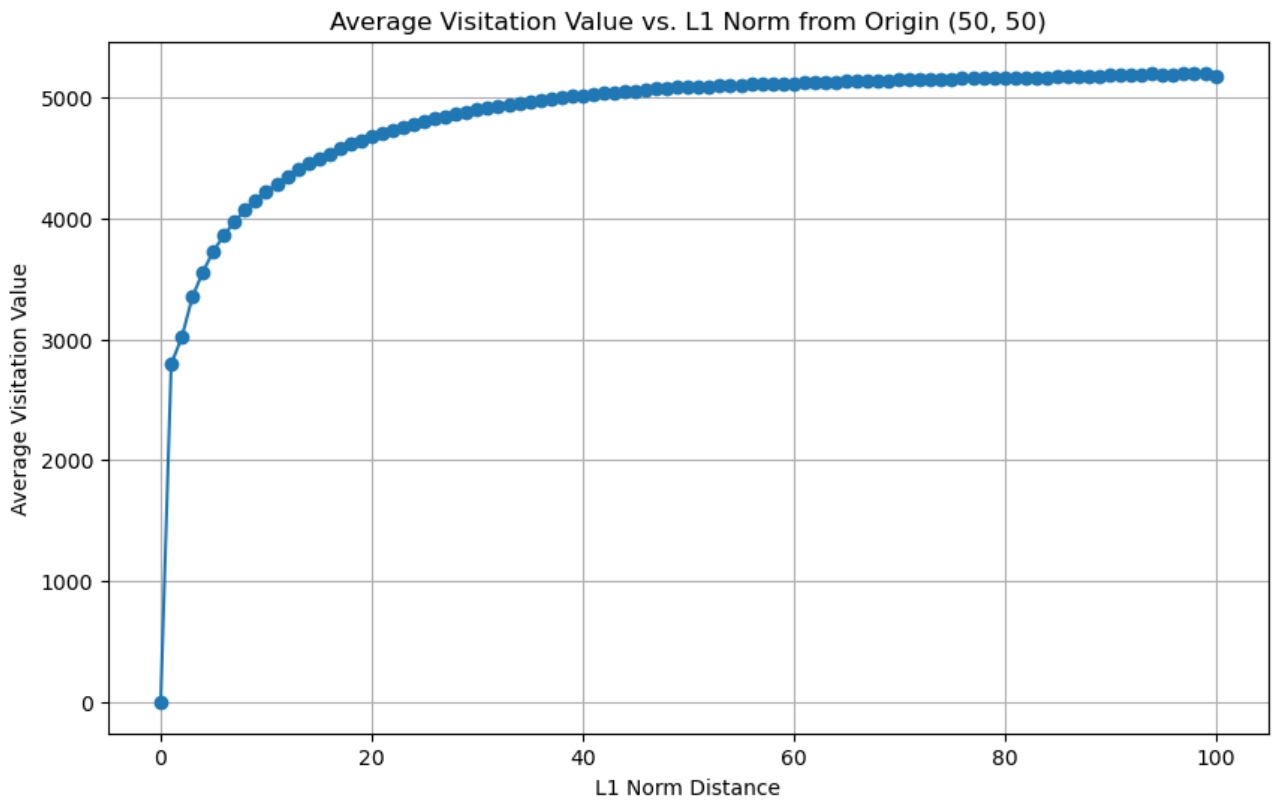
Figure 4: Probability variation with Euclidean distance.

before the target node, if the target nodes are closer to the origin. Upon plotting this with respect to distance, a monotonically increasing concave curve is obtained in which, initially, an increase in distance causes an increase in cost. However, after about a distance of 30 from the origin the cost seems to plateau just above 5000 giving a combined average of 4999.5 which is just less than $5000 = n/2$ as $n = 10000$ in 100×100 grid.

Peculiarly, rDFS's DFS-numbering vs distance curve too exhibits the same phenomena. This similarity doesn't stop there either; like the heat map, the probability heatmaps for both rDFS and random walk look quite similar too except for the minor difference being the ball around random walk origin being denser (higher visitation probabilities) than that of rDFS. Upon plotting the probabilities with respect to distance, however, it is possible to observe that the random walk probabilities gradually decrease for nodes at a distance from origin till 40, and then plateaus around just below 0.5. Once again, the similarly generated probability curve for rDFS too exhibits the same property.

At a high level, it is possible to interpret these phenomena. The only major difference between simple random walk and other traversal methods is that the random walk takes much higher steps (hitting time for a random walk on grid is $O(n^3)^4$) compared to that of BFS/DFS (O(n)). Much of this can be attributed to the amount of steps the walk spends within it's previously discovered node set before hitting a new node. However, in the Alpern-Lidbetter setting, the cost of traversal within previously explored nodes is negligible. Hence, cost is only incurred when the walk hits new nodes that are not hider. Owing to the nature of an unbiased random walk, it hits all nodes at a distance d from source in all directions in a uniform manner. This fits neatly with the expanding search paradigm explored in the aforementioned papers, as the random walk can indeed traverse within previously discovered nodes and "restart" the search into new nodes from the neighbourhood of any previously explored nodes.

In the case of rDFS, conditional probabilistic arguments can be used to show why nodes closer to the origin have higher probability of being visited during a search. Over many runs, the random choosing from the four neighbours around origin, and similarly next and so on, ensures this decreasing probability too applies uniformly to nodes in all directions.



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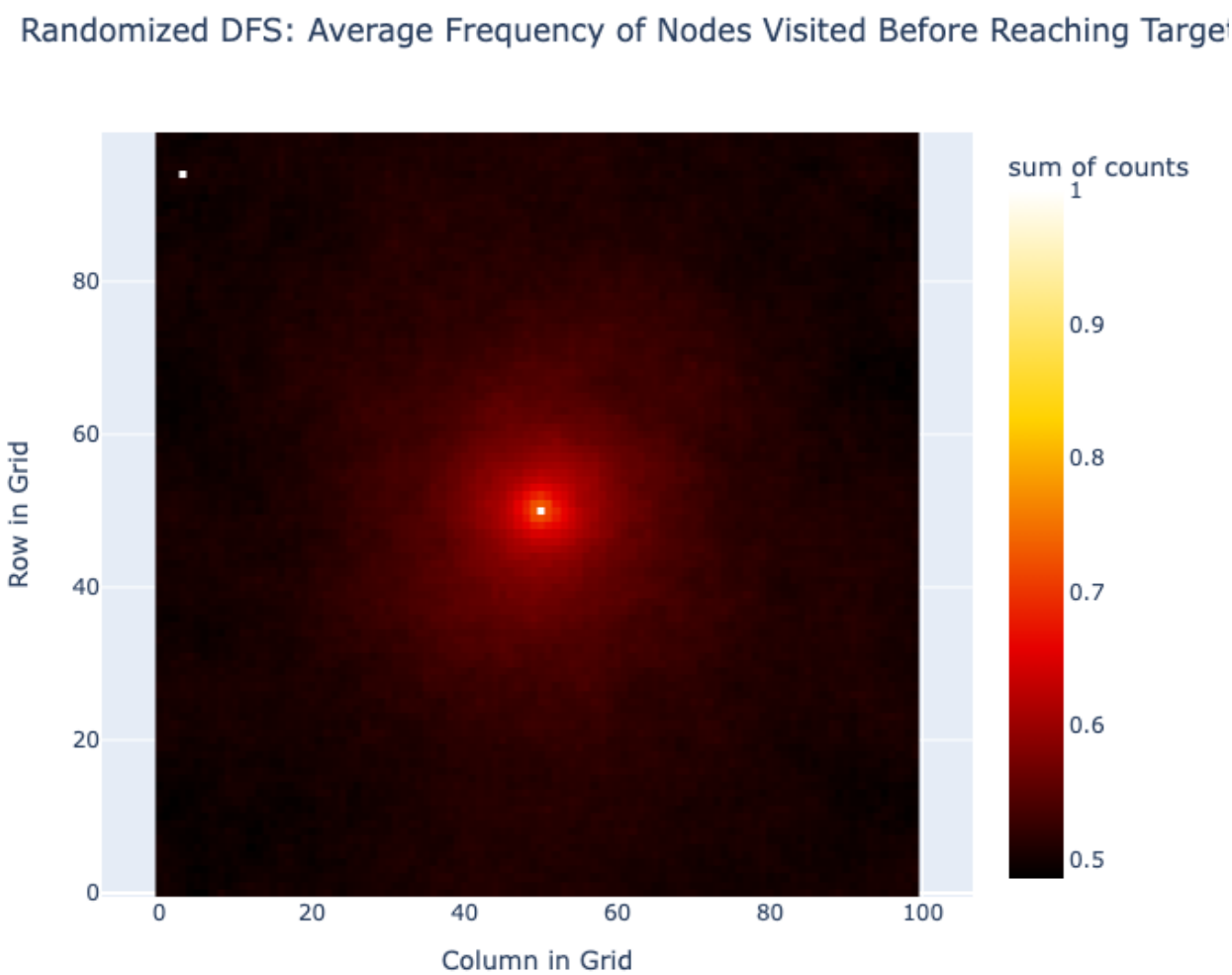

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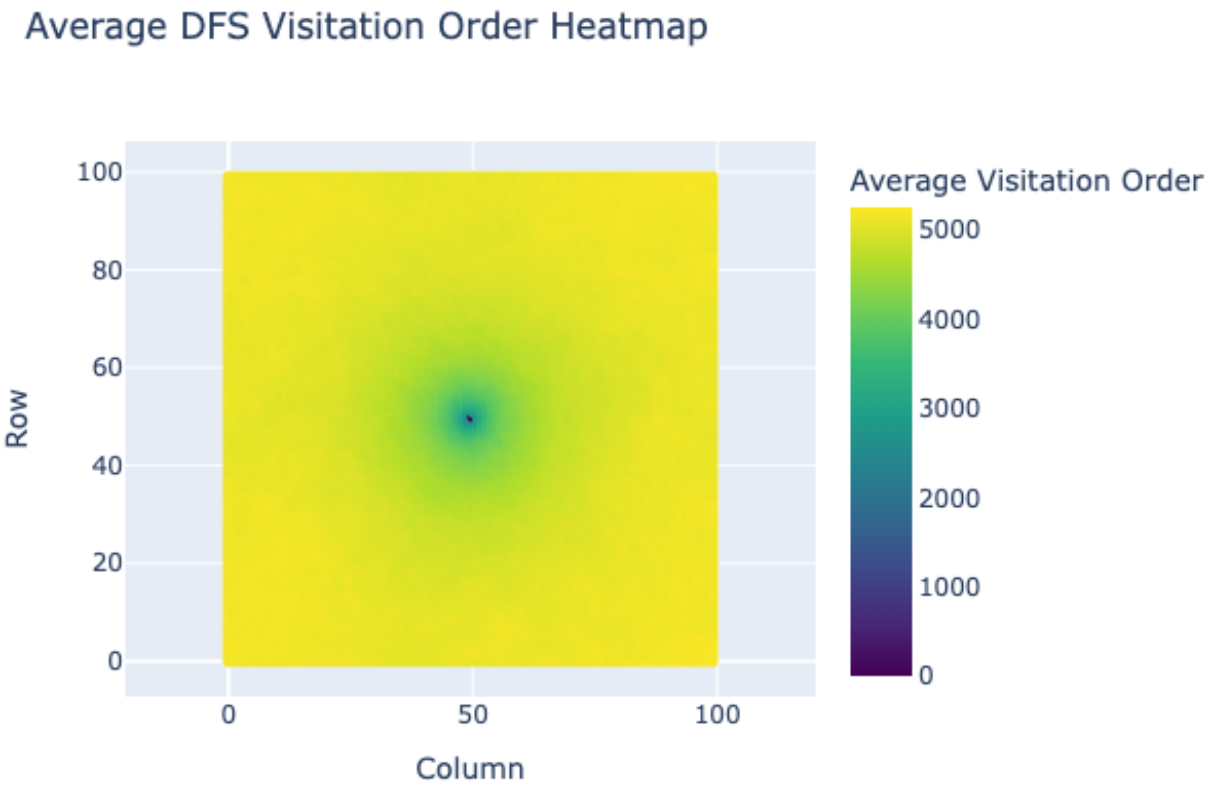

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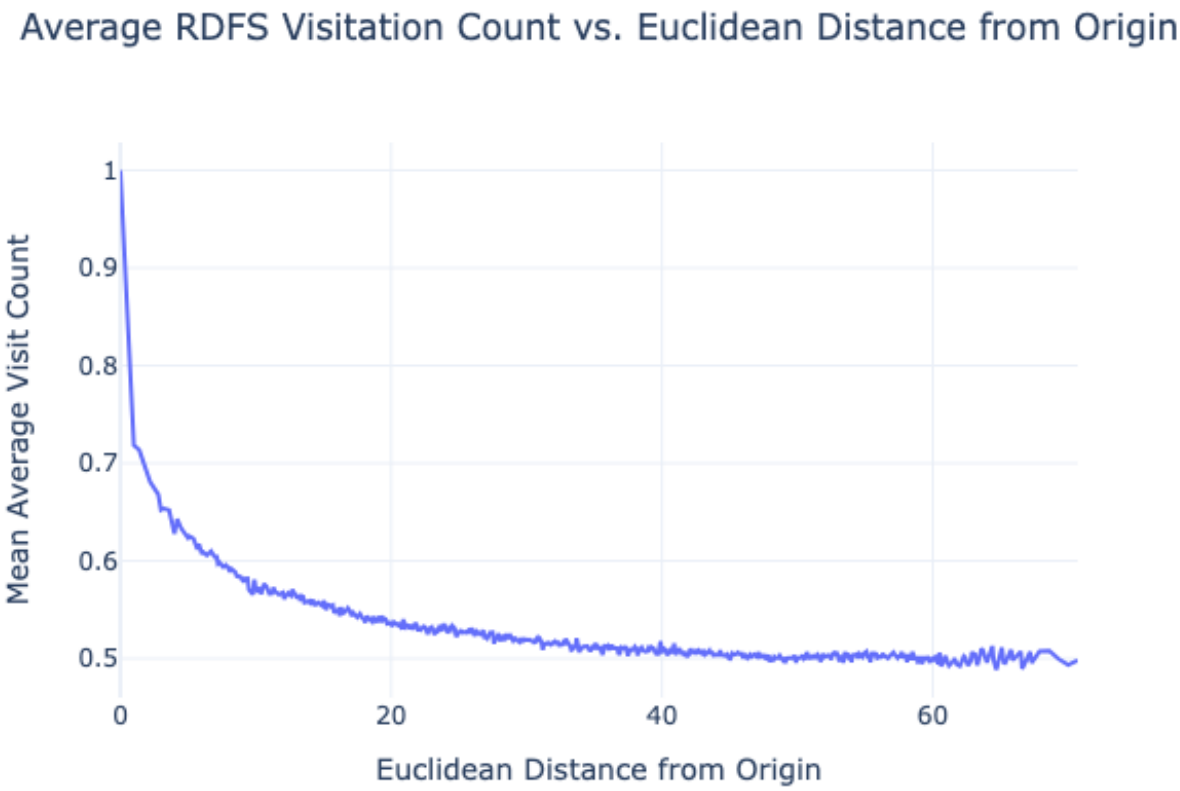


Figure 8

Average Number of Steps to Reach Each Target Node:

Strategy	BFS	DFS	Randomized DFS
Target Node			
(3, 94)	9827.0	392.0	5148.271
(4, 3)	9896.0	494.0	5231.746
(11, 27)	7058.0	1217.0	5128.621
(28, 17)	5916.0	2856.0	5046.980
(35, 31)	2283.0	3597.0	4959.993
(69, 11)	6484.0	6959.0	5053.308
(75, 54)	1737.0	7522.0	4864.030
(81, 14)	7837.0	8168.0	5034.454
(86, 94)	9226.0	8693.0	5345.036
(94, 13)	9292.0	9482.0	5105.399

Figure 9

Without making use of any information gained from the network exploration, random walk / rDFS seems to be the best strategy for exploration and searching. The onset of these experiments suggested the possibility of random walk giving a definitive upper-bound or lower bound over the randomized depth first search; however, as the following analysis reveal this is not to be the case.

Random Walks and RDFS: Differences and Possible Explanations

The L1 and L2 distance-wise average cost to hit nodes was plotted for both randomized depth first search and the random walk on a 100 by 100 toroidal grid. Instead of any one process having a clearly dominating behavior, it was revealed that the cost of randomized depth first search remains higher than the random walk initially, rising rapidly to 5000. Meanwhile, the random walk finds nodes closer to the origin at a lower cost. At $d/2$, with d being the maximum euclidean distance starting from origin ($d = app.70$ and $d/2 = app.35$ in our setting), there's a crossover between the costs of these processes with the random walk now seeming to take on a higher cost to find nodes at a distance more than $d/2$.

This difference in behaviour can be attributed to 2 properties unique to the walk and depth first search respectively arising from the fact that the random walk can have self intersecting path while the randomized depth first search cannot. First, the random walk has a strong local tendency with the possibility of self intersection, allowing the process to visit nodes lying at the

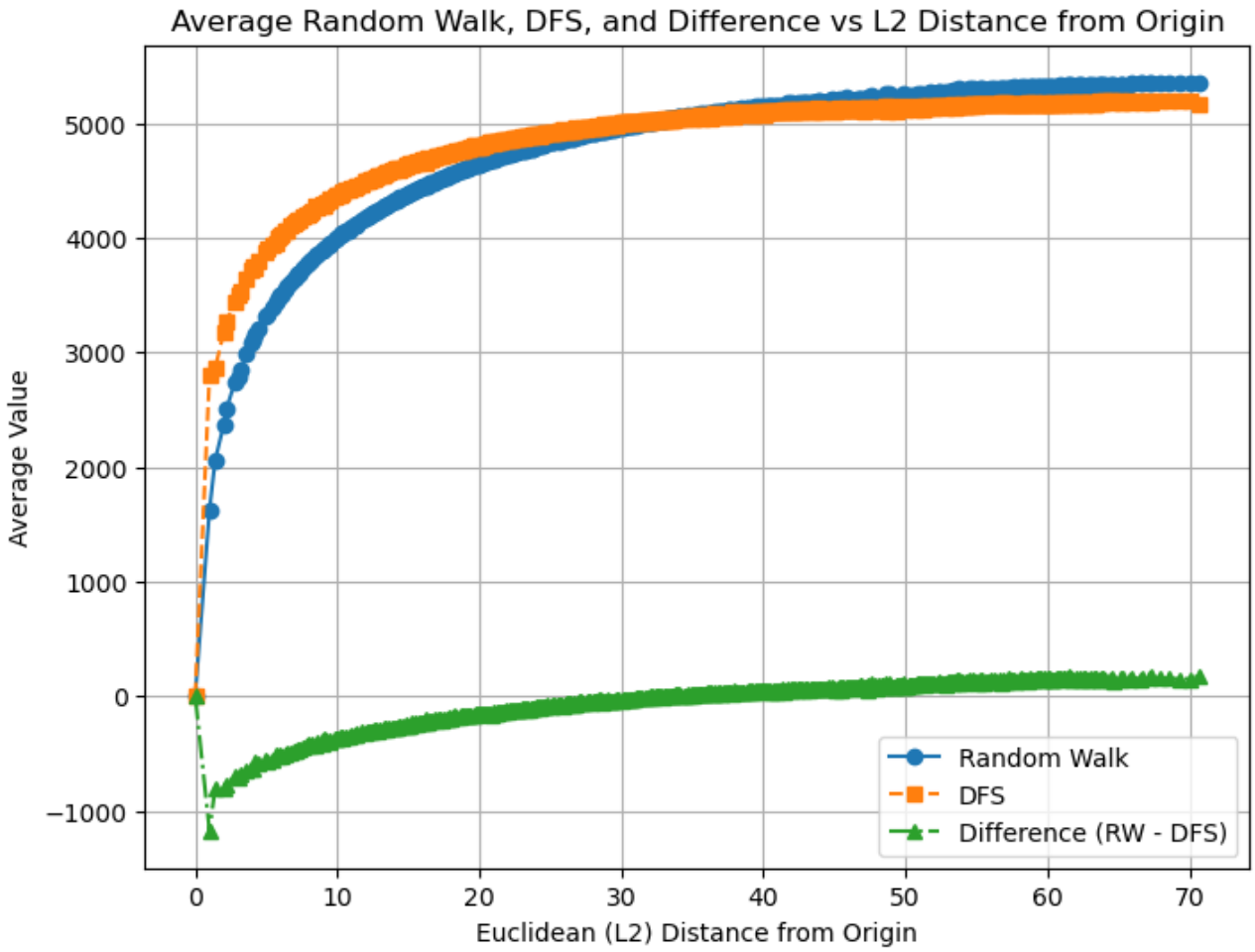


Figure 10

same distance from origin in different directions with a higher probability than nodes that may lie at greater distance (this is only relative to rDFS). On the other hand rDFS has a tendency to create a boundary with each step, creating a directional bias in the long run pushing the process to go further along a particular direction with nodes at a closer distance at a different direction being left behind with greater probability. Despite these distance wise differences, the average cost of both the random walk and randomized depth first search on a n node toroidal grid come out to be exactly $\frac{n-1}{2}$.

Symmetry of the Walk:

Despite these differences in behaviour, random walk offers the most accessible way to analyse probabilistic processes in regular graphs. To this end, the first problem to tackle is the question of symmetry: whether the walk is symmetric along the L1 or L2 circles of the toroidal grid.

This will be helpful determining along what lines further analysis can be conducted. And so, the average cost along l1 and l2 norms was looked into; furthermore, the average first visit times was plotted for nodes with the same l1 distance but varying l2 distance. It was possible to see that nodes with higher l2 distance had a higher cost while nodes with a lower l2 distance but with the same l1 distance had a lower cost. No similar behaviour can be observed on nodes with same l2 distance and different l1 distances.

It is possible to make sense of this behaviour by understanding the geometry of the L1 circle: Consider a point (x, y) in the first quadrant. Its Manhattan distance (L_1) from the origin is

$$d_{L_1} = x + y,$$

and its Euclidean distance (L_2) is

$$r = \sqrt{x^2 + y^2}.$$

Now, fix a Manhattan distance d; that is, consider all points satisfying

$$x + y = d.$$

We can parameterize these points by writing $x = d - y$ with $y \in [0, d]$. Their Euclidean distance is

$$r^2 = (d - y)^2 + y^2 = d^2 - 2dy + 2y^2.$$

To find the extrema of r for a fixed d, differentiate with respect to y:

$$\frac{d}{dy} r^2 = -2d + 4y.$$

Setting this derivative equal to zero gives:

$$-2d + 4y = 0 \implies y = \frac{d}{2}.$$

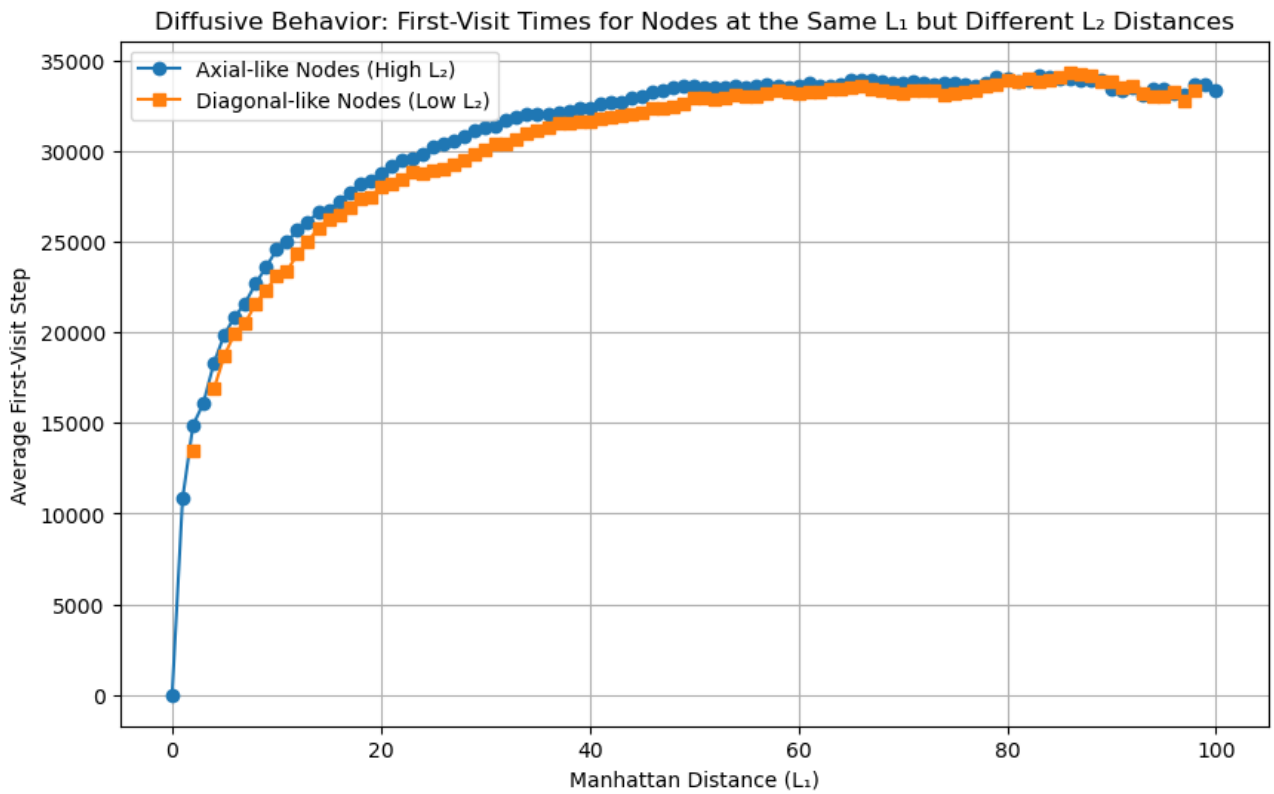


Figure 11

At $y = \frac{d}{2}$, we have $x = \frac{d}{2}$ and

$$r_{\min} = \sqrt{\left(\frac{d}{2}\right)^2 + \left(\frac{d}{2}\right)^2} = \frac{d}{\sqrt{2}}.$$

At the endpoints of the segment (e.g., $y = 0$ or $y = d$), one obtains:

$$r_{\max} = d.$$

Thus, although every point on the L_1 "circle" $x + y = d$ shares the same Manhattan distance, their Euclidean distances range from $\frac{d}{\sqrt{2}}$ (for the diagonal or "corner" points) to d (for the axial points).

Spectral Theory and Approaches:

With all this established, the focus was turned now to study the flow of random walks with respect to the type of graphs on which the walk takes place, thereby trying to access the

rich mathematical theory associated with spectral analysis of random walks and probability on trees and networks. With the walk itself free of any bias except for a fixed starting point, the most important characterization becomes that of the idea of "resistence" provided by the underlying graph. In this sphere, Piyush, in his write-up "Random Walks with Volume Cost"[5], has provided the following theory which is greatly effective in capturing the many behaviours of the process in hand.

Preliminaries

Setup: Let $G = (V, E)$ be an undirected connected simple graph. Given a random walk W, possibly non-Markovian, on the vertices of G, and started at a fixed vertex $u \in V$, we define the cost $\text{cost}_W(u, v)$ to be the number of *distinct* vertices of G visited by W before v is visited for the first time. Thus, $\text{cost}_W(u, u) = 0$, and $|V| - 1 \geq \text{cost}_W(u, v) > 0$ for every $v \neq u$. Our goal is to understand the expected maximum cost:

$$\max_{v \in V} E[\text{cost}_W(u, v)] \tag{1}$$

for various random walks W and starting states u. Two specific random walks of interest are:

1. The randomized depth first search (**RDFS**).
2. The standard Markovian random walk on a (positively) weighted undirected graph.

Standard Random Walks

Now, we equip the undirected graph $G = (V, E)$ with a positive edge weight function w. We assume that G is connected. As usual, we also view G as a network of resistors, in which each edge e has resistance $r(e) := 1/w(e)$. The weight function also defines a natural irreducible Markov chain P_w on V, with transition matrix:

$$P_w(u, v) = \begin{cases} \frac{w(\{u,v\})}{\sum_{x:\{u,x\} \in E} w(\{u,x\})} & \text{if } \{u, v\} \in E, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

We also define the unnormalized Laplacian $L_w := D_w - W$, where D_w is a diagonal matrix in which for each $v \in V$, $D(v,v) = \sum_{x:\{x,v\}\in E} w(\{u,x\})$, and W is a matrix where:

$$W(u,v) = \begin{cases} 0 & \text{if } \{u,v\} \notin E, \\ w(\{u,v\}) & \text{otherwise.} \end{cases} \tag{3}$$

Since G is connected and has positive weights, L is a symmetric matrix with rank deficiency exactly 1, and its null space is spanned by the vector 1.

Effective Resistance

Definition 2.1 (Effective Resistance). Given a simple, undirected, connected graph $G = (V, E)$ with positive edge weights w as above, the *effective resistance* between vertices u and v in G is defined to be the voltage difference between u and v when a current source with input of one unit of current is attached to u and a current sink with input of one unit of current is attached to v. Formally, it is the (uniquely determined) value $x_u - x_v$ in any solution to the linear system:

$$L_w x = (e_u - e_v), \tag{4}$$

where e_t for $t \in V$ denotes the standard unit basis vector in $R^{|V|}$ in which the only non-zero entry corresponds to the vertex t. The effective resistance is guaranteed to be positive.

Key Results

We denote the effective resistance between u and v as $R^G_{\text{eff}}(u,v)$, and drop the explicit dependence on the graph and weights when these are clear from the context. Using the Moore-Penrose pseudoinverse L_w^+ of L_w, we can write:

$$R^G_{\text{eff}}(u,v) = (e_u - e_v)^T L_w^+ (e_u - e_v)^6. \tag{5}$$

It is now possible to understand the cost $E[\text{cost}_W(u,v)]$ under these definitions where u and v are vertices in V.

proposition:

Let u and v be distinct vertices in an undirected, simple, connected graph G = (V, E) with positive edge weights w. Consider a random walk p_w defined as before. Consider an electric circuit in which each edge e of G has resistence $1/w(e)$ in which to each vertex $x \neq v$ a current source providing current $1_{R^G_{\text{eff}}(x,v)}$ is attached; to v a current source drawing current $\sum_{x \in V, x \neq v} 1/R^G_{\text{eff}}(x,v)$ is attached. Then, expectation of cost $E[\text{cost}_W(u,v)$ equals the voltage difference between u and v in the circuit.

proof of the proposition:

Fix a vertex $x \neq v$ and consider the function $h_x : V \setminus \{x, v\} \to [0,1]$ such that for any vertex $y \in V, h_x(y)$ is the probability that the walk starting at y visits x before v. Through linearity of expectation,

$$E[\text{cost}_W(u,v)] = \sum_{x \in V \setminus \{v\}} h_x(u) \tag{6}$$

The function h_x is termed the harmonic, with conditions $h_x(x) = 1$, $h_x(v) = 0$ and that all entries corresponding to $L_w h$ except those corresponding to x and v are zero. If L^+ is the moore-penrose pseudo-inverse of L_w, it follows from the definition of effective resistence that for any $x \neq v$ and any vertex $u \in v$

$$h_x(u) = \frac{1}{R^G_{\text{eff}}(x,v)} (e_u - e_v)^T L^+ (e_x - e_v) \tag{7}$$

Upon substituing which into the previous equation,

$$E[\text{cost}_W(u,v)] = (e_u - e_v)^T L^+ \sum_{x \in V \setminus \{v\}} \left(\frac{e_x - e_v}{R^G_{\text{eff}}(x,v)}\right) \tag{8}$$

Simulation of this theory has proved to be quite effective as it precisely agrees with the earlier experimental costs derived. A closed form solution for the cost for line/path graphs was provided in the said write-up; however, there are significant obstacles to deriving such a solution for the settting in hand of a toroidal grid, associated with characterizing the resistence over

multiple paths, connected into a grid. While this is a promising approach towards the conjecture in hand, another approach that's been taken concerns finding monotonicity properties and establishing $n/2$ as an upperbound.

Matrix Properties and Cost Computation:

A major setback with directly applying the above stated approach to calculating the cost of points is the fact that computation of cost for n points takes upto $O(n^4)$ time. And so, the search for more effective characterizations of cost is the next sensible step. To this end, the characterizing matrices (i.e, adjacency, laplacian, and laplacian inverse) of a graph were examined for possible structure. One can clearly see the existence of possible strucutre in these matrices.

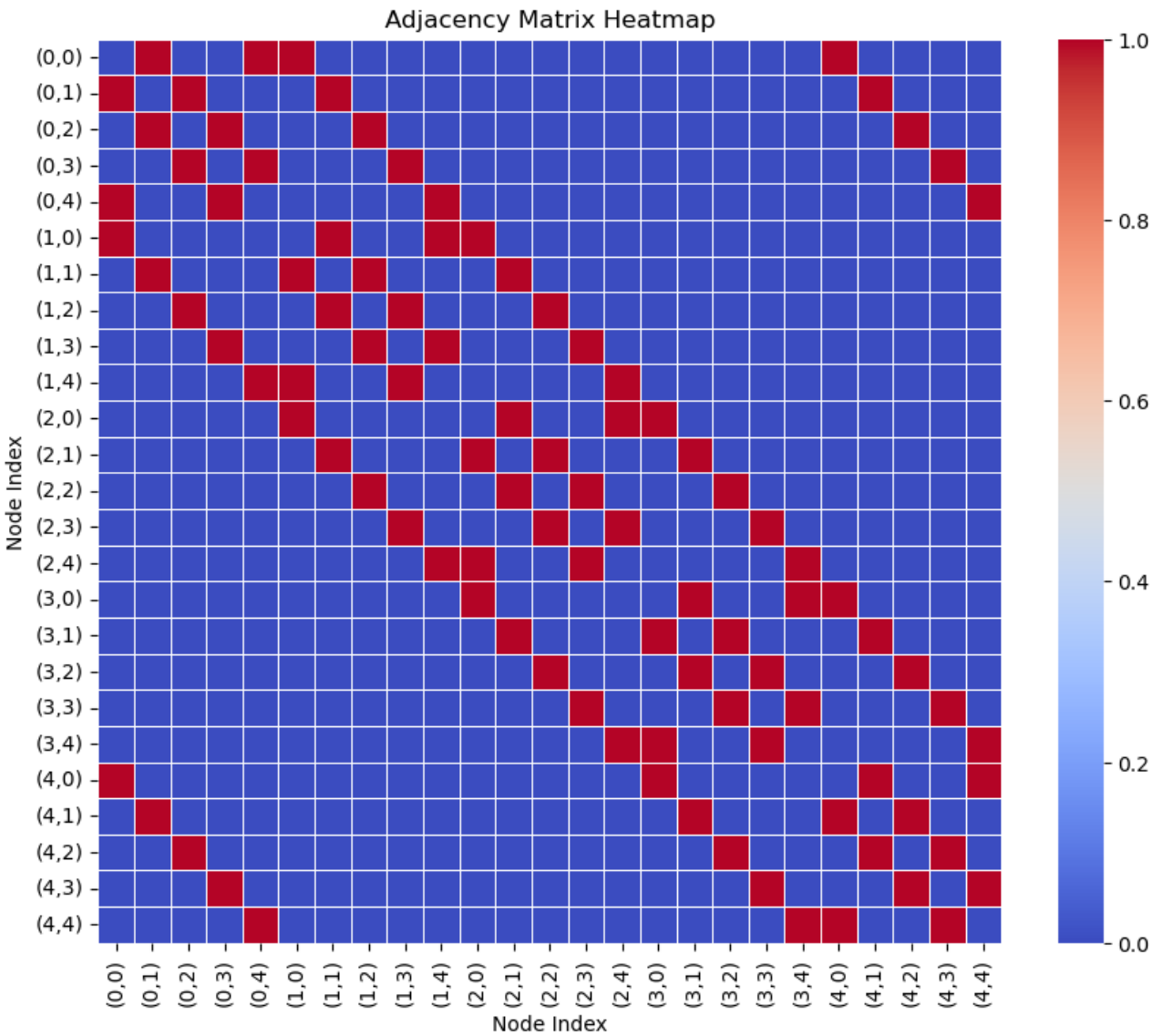
First found is a method to compute the expected hitting cost $E[\text{cost}_W(u, v)]$ on an $n \times n$ toroidal grid (and more generally on sparse graphs) using a single sparse factorization plus multiple cheap back-substitutions, rather than $O(N)$ factorizations.

Efficient Computation of $E[\text{cost}_W(u, v)]$ via a Single Sparse LU Factorization

Baseline Approaches and Their Cost

Dense pseudoinverse. Forming $L_w^+ \in^{N\times N}$ via SVD or dense Gaussian elimination requires $O(N^3)$ time and $O(N^2)$ memory. Once available, each $h_x(u)$ is an $O(N)$ inner product, but the dominant cost remains $O(N^3)$.

Repeated sparse solves. Remove row and column v to form the nonsingular $(N-1)\times(N-1)$ minor $L_{-\{v\}}$. For each $x \neq v$, factor $L_{-\{v\}}$ (cost $O(N^{1.5})$ for a 2D grid) or apply a new sparse-LU, solve for the xth column of $L_{-\{v\}}^{-1}$, and read off the diagonal entry. Repeating $N-1$ times yields $O(N \cdot N^{1.5}) = O(N^{2.5})$.


Adjacency Matrix Heatmap
Node Index
Node Index
(0,0)
(0,1)
(0,2)
(0,3)
(0,4)
(1,0)
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(1,2)
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(4,3)
(4,4)
1.0
0.8
0.6
0.4
0.2
0.0

Figure 12

All harmonic-function solves

$$L_{-\{v\}}\, y^{(x)} = e_x \quad (x \neq v),$$

and the final voltage solve

$$L_{-\{v\}}\, \phi_{-\{v\}} = \sum_{x \neq v} \frac{e_x}{R_{\text{eff}}(x, v)}$$

share the *same* coefficient matrix $L_{-\{v\}}$. By linearity (superposition in circuits), we can factor once and reuse that factor for every right-hand side.

Single-Factor Method

We replace $N-1$ factorizations by a single sparse LU (or Cholesky) on $L_{-\{v\}}$, plus N back-substitutions:

- **Build** the full Laplacian $L_w \in^{N \times N}$ of the toroidal grid.
- **Form** the $(N-1)$-by-$(N-1)$ minor

$$L_{-\{v\}} = L_w\Big[V \setminus \{v\},\, V \setminus \{v\}\Big].$$

- **Factorize** $L_{-\{v\}} = LU$ once via a sparse solver in $O(N^{1.5})$.
- **Compute effective resistances.** For each $x \neq v$, solve $L_{-\{v\}}\, y^{(x)} = e_x$ by forward/back substitution in the existing LU. Then $R_{\text{eff}}(x, v) = y_x^{(x)}$. This is $N-1$ solves at $O(N)$ each $\implies O(N^2)$.
- **Assemble current vector.** Define $b \in^N$ by $b_x = 1/R_{\text{eff}}(x, v)$ for $x \neq v$, and $b_v = -\sum_{x \neq v} b_x$.
- **Voltage solve.** Restrict b to $b_{-\{v\}} \in^{N-1}$ and solve

$$L_{-\{v\}}\, \phi_{-\{v\}} = b_{-\{v\}}$$

  by one more back-substitution. Set $\phi_v = 0$. Then $\phi_u = E[\text{cost}_W(u, v)]$.

Complexity Comparison

Method	# Factorizations	# Back-solves	Asymptotic Cost (2D grid)
Dense pseudoinverse	0	0	$O(N^3)$
Repeated sparse LU	N	0	$O(N^{2.5})$
Single-factor (this work)	**1**	N	$O(N^{1.5} + N^2)$

Table 1: Cost comparison for computing $E[\mathrm{cost}_W(u, v)]$.

Since $N = n^2$, the leading terms scale as

$$\text{Single-factor: } O(n^3 + n^4), \quad \text{Repeated: } O(n^5), \quad \text{Dense: } O(n^6).$$

Extensions and Remarks

- **Near-linear solvers.** Replace the sparse LU by a nearly-linear-time Laplacian solver (e.g. Koutis–Miller–Peng) to drive the factor step to $O(N \operatorname{polylog} N)$.
- **Iterative back-solves.** If only approximate voltages or resistances are needed, one can solve $L_{-\{v\}}\, y = e_x$ iteratively, potentially reducing the back-solve cost below $O(N)$ per right-hand side.
- **Other graphs.** Although we focused on the toroidal grid for concreteness, the same idea applies to any sparse graph where a single factorization is cheap relative to repeated refactorizations.

In summary, the single-factor method leverages the superposition principle of electrical networks to collapse N costly factorizations into one, yielding dramatic savings both asymptotically and in practice on large sparse graphs.

Spectral (2D DFT) Computation of Effective Resistances

We consider the special case of the $n \times n$ toroidal grid graph $G = C_n \times C_n$, where C_n denotes the cycle graph on n vertices. The full translational symmetry of this graph enables us to compute the effective resistance $R_{\text{eff}}(x, v)$ in closed form using the two-dimensional discrete

Fourier transform (2D DFT).

Circulant Structure and Diagonalization

- The Laplacian matrix of the cycle graph C_n is a circulant matrix $L_{C_n} \in R^{n\times n}$. It is defined as

$$(L_{C_n})_{i,j} = \begin{cases} 2 & \text{if } i = j, \\ -1 & \text{if } i \equiv j \pm 1 \mod n, \\ 0 & \text{otherwise.} \end{cases}$$

- The Laplacian of the toroidal grid $G = C_n \times C_n$ is given by the Kronecker sum

$$L = L_{C_n} \otimes I_n + I_n \otimes L_{C_n},$$

  where $\otimes$ denotes the Kronecker product and I_n is the $n \times n$ identity matrix.

- Any circulant matrix is diagonalized by the discrete Fourier transform (DFT) matrix $F_n \in C^{n\times n}$, with entries

$$(F_n)_{jk} = \frac{1}{\sqrt{n}} e^{-2\pi ijk/n}, \quad 0 \le j, k < n.$$

- Thus, the Laplacian L is diagonalized by $F_n \otimes F_n$:

$$L = (F_n \otimes F_n)^* \cdot \mathrm{diag}(\lambda_{p,q})_{0\le p,q<n} \cdot (F_n \otimes F_n),$$

  where $\lambda_{p,q}$ are the eigenvalues given by

$$\lambda_{p,q} = 4 - 2\cos\left(\frac{2\pi p}{n}\right) - 2\cos\left(\frac{2\pi q}{n}\right).$$

Spectral Formula for Effective Resistance $R_{\text{eff}}(x, v)$

- Let $L_{-\{v\}}$ denote the Laplacian matrix with the row and column corresponding to a fixed vertex v removed. Its inverse, $L^{-1}_{-\{v\}}$, is the Green's function $G_{x,y}$ of the graph grounded

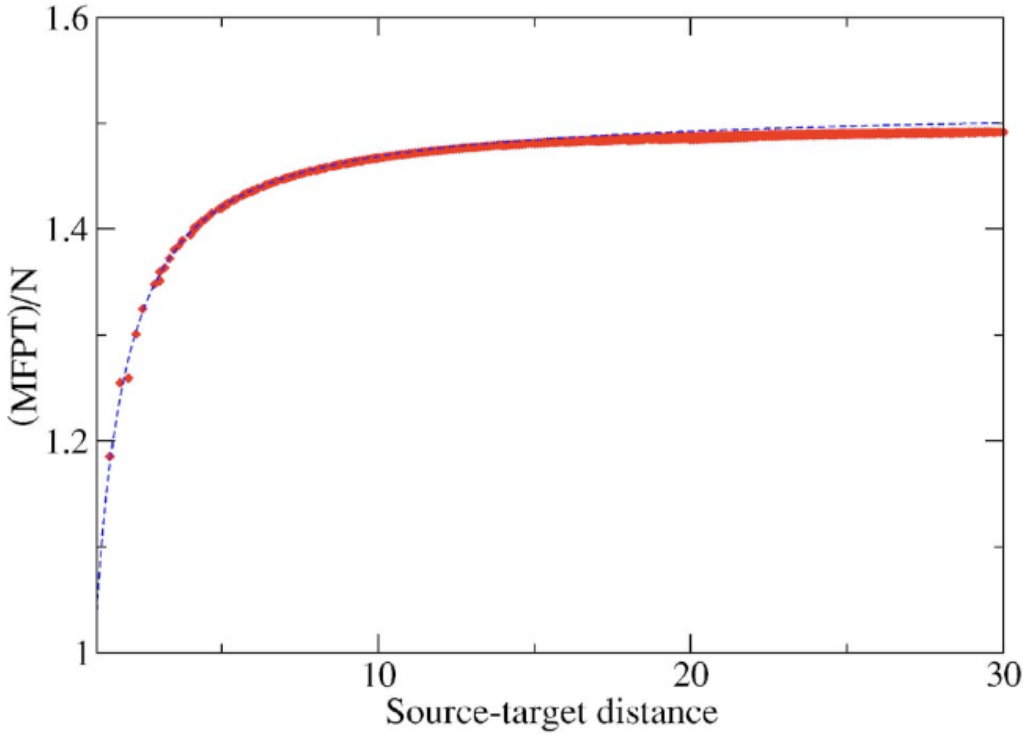

FIG. 2. (Color online) Three dimensions: Influence of the distance between the source and the target. Red diamonds: simulations; blue dashed line: evaluation of the FPT with $H=G_0$. The

Figure 13

at vertex v.

- The effective resistance between vertices x and v is given by

$$R_{\text{eff}}(x,v) = (L^{-1}_{-\{v\}})_{x,x}.$$

- Using Parseval's identity and the DFT expansion, we get:

$$R_{\text{eff}}(x,v) = \frac{1}{n^2} \sum_{\substack{0\le p,q<n \\ (p,q)\neq(0,0)}} \frac{1-\cos\left(\frac{2\pi}{n}\left[p(i_x-i_v)+q(j_x-j_v)\right]\right)}{\lambda_{p,q}},$$

  where (i_x, j_x) and (i_v, j_v) are the grid coordinates of vertices x and v respectively.

This summation yields the so-called lattice Green's function, a well-studied object in probability and mathematical physics, used to explain resistance and hitting probabilities.

The Lattice Green's Function on the Toroidal Grid

The Green's function $G_{u,v}$ is the (u, v) entry of the pseudoinverse L^+ of the Laplacian matrix L. For any connected graph,

$$R_{\text{eff}}(u, v) = G_{u,u} + G_{v,v} - 2G_{u,v}, \quad \sum_{x \neq v} h_x(u) = R_{\text{eff}}(u, v),$$

where $h_x(u)$ denotes the expected hitting time from u to x.

On the toroidal grid, the translational symmetry implies:

- The Green's function depends only on the lattice displacement $(\Delta i, \Delta j)$ between u and v.
- The 2D DFT leads to explicit double-sum (and in the limit, double-integral) representations.
- Finite-size effects lead to a flattening of $G(r)$ for $r \to n/2$, due to wraparound.

Asymptotic Behavior via the Lattice Green's Function

The trace of the Laplacian pseudoinverse is

$$\sum_u G_{u,u} = \sum_{\substack{0 \leq p,q < n \\ (p,q) \neq (0,0)}} \frac{1}{\lambda_{p,q}},$$

which, in the limit as $n \to \infty$, approaches the classical lattice Green's function on Z^2:

$$\frac{1}{\pi^2} \int_0^\pi \int_0^\pi \frac{1}{2 - \cos x - \cos y} \, dx \, dy.$$

This integral diverges logarithmically, implying that on a finite torus:

$$G(r) = G_{u,v}(r = \|u - v\|) = \frac{1}{2\pi} \log r + C + o(1),$$

leading to the observed $\log n$ growth in resistance and discovery cost[10].

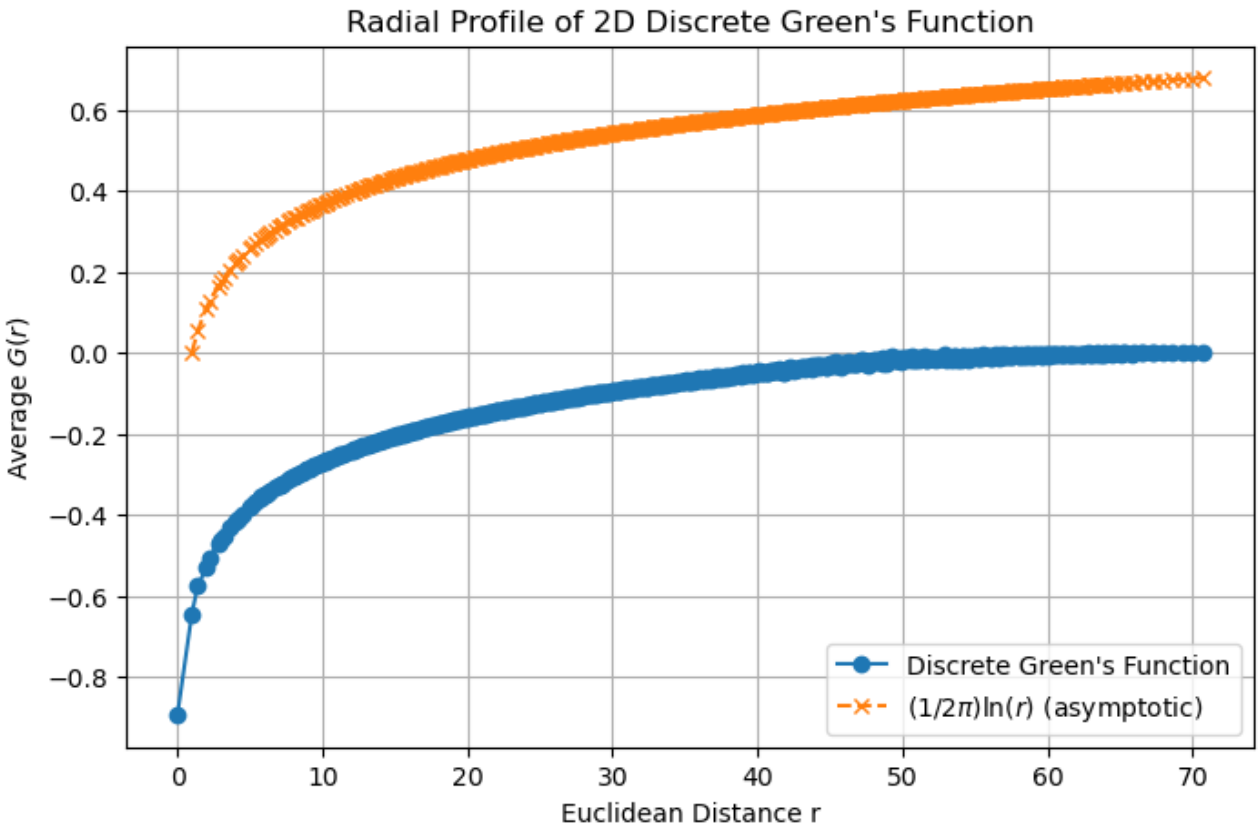

Figure 14

On the torus the only thing closed-form Green's function "sees" is the relative displacement $(i_1 - i_2, j_1 - j_2)$—there simply aren't any structural wrinkles (bottlenecks, clusters, high-degree hubs...) to show up. In contrast, on a highly asymmetric graph like a barbell: Two cliques joined by a bridge breaks vertex-transitivity, so $G^+_{u,v}$ (and hence resistances, hitting-probabilities, discovery-costs) will depend on which clique we start in, how we approach the bridge, etc. No single "distance from the origin" coordinate can summarize that structure, and there is no global Fourier diagonalization to collapse L^+ down to a one-variable function.

The DFT sum

$$\frac{1}{n^2} \sum_{(p,q)\neq(0,0)} \frac{2-2\cos\big(p\Delta_i + q\Delta_j\big)}{\lambda_{p,q}}$$

is exactly a function of the lattice displacement (Δ_i, Δ_j). It encodes how "far away" one is on the grid, nothing more. These calculations are performed and stated in the paper "Random Walks and Brownian motion: A method of computation for first-passage times and related quantities in confined geometries" with respect to the mean first passage times. However, we observe that our notion of cost is highly related to MFPT from the initial experiments, with a

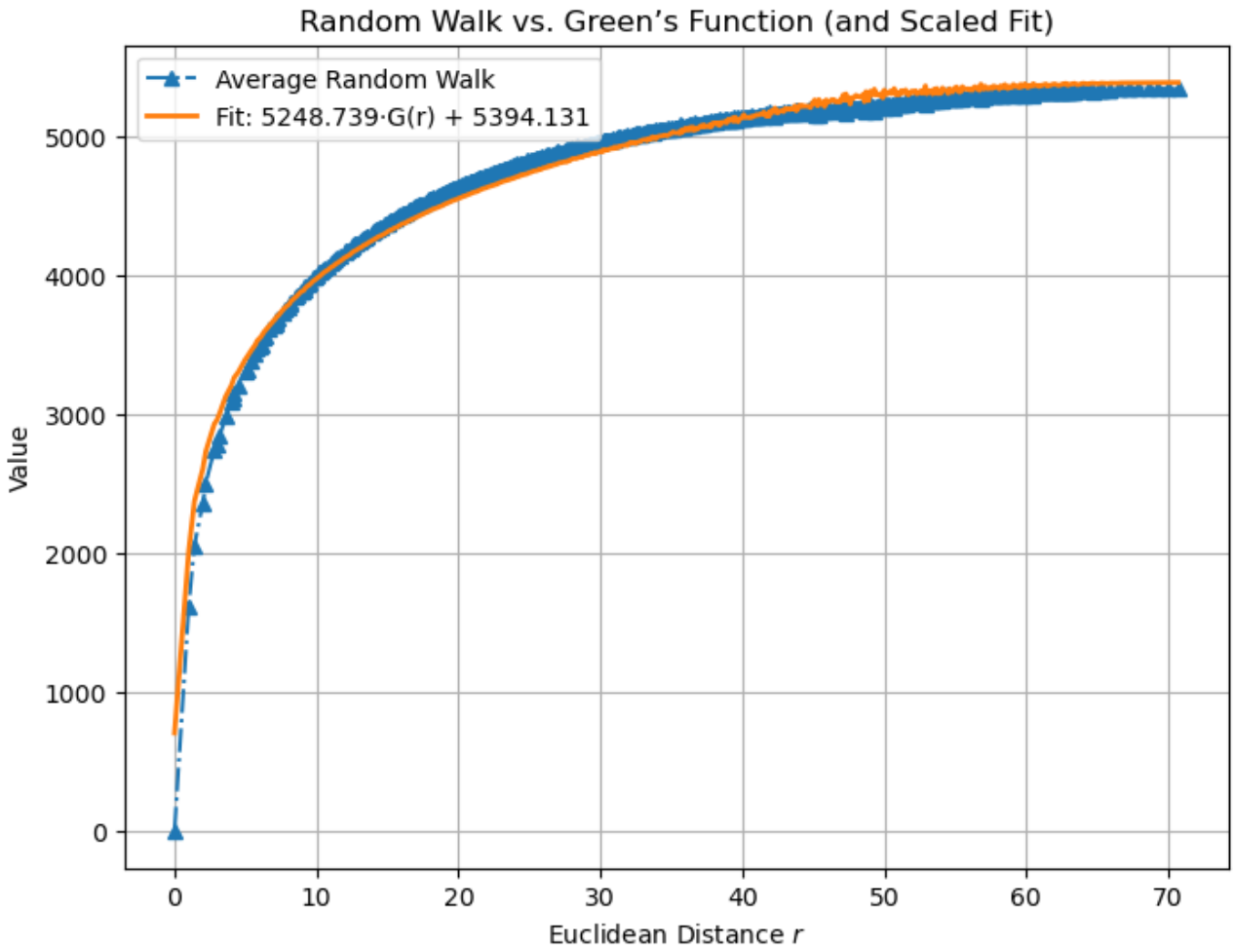

Figure 15

one to one correspondence between the curves save only for a scaling factor. In fact, the distance vs MFPT plot provided in the stated paper matches our cost curve exactly, showcasing how distance driven green's function is the only governing factor.

It is possible to see in figure 14, that the radial profile of the lattice green function looks rather similar to the plots we've been dealing with in-terms of cost of the random walk. In fact, the overlay plot—with a best-fit line $C1G(r) + C2$ matches the empirical random-walk curve almost point-for-point, establishing a linear relationship between the green's function and our cost. On the finite torus of side n, distances "wrap around," so the maximum meaningful separation is $n/2$. Beyond that we get back-side interactions from the periodic boundary.

The emergence of green function like behavior hints at approaches for generalizing this behaviour as the same principle applies to any graph whose adjacency (or Laplacian) lives in the

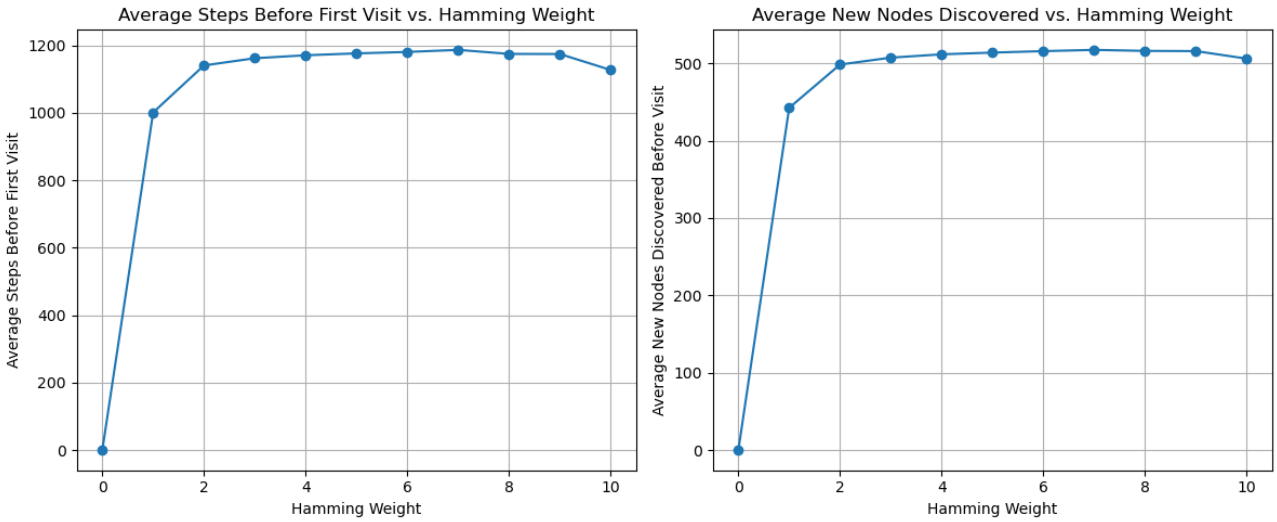

Figure 16

group algebra of a finite abelian group. Examples include:

- Cycles (Z_n-circulant 1D DFT).
- Hypercubes ($(Z_2)^d$-Cayley Walsh–Hadamard transform) – this also explains similar behaviour observed with hypercube simulation with random walks.
- Cartesian products of cycles, hypercubes, etc.

More generally, any Cayley graph of an abelian group. All of these admit an explicit "Fourier" or "character-sum" formula for L^+. Outside of those highly symmetric families—most irregular graphs or non-abelian Cayley graphs—no such closed form is available.

With the question of cost behavior now settled, it now only remains to understand why such behaviour occurs around n-1/2. The following section takes up this question with an attempted pairing argument.

Unpacking $\frac{n-1}{2}$

In order to try and see why $n - 1/2$ arises naturally on regular graphs, consider the spectrum of connected graphs with respect to our notion of cost. On one end of the spectrum, are line graphs – with only one path between nodes– where cost can be as much as $n - 1$ even though the average comes out to be $n - 1/2$, on the other end of the spectrum are fully connected

graphs with every node connected to every other node; here, the question of expected cost collapses to the expected position in a random permutation of n objects which is

$$\frac{n-1}{2}$$

which is also the lower bound for our notion of cost for any type of graph as fully connected graph have maximal randomness (in the sense the visitation order of the walk is truly random). Intiutively, it is possible to observe that regular graphs lie closer to fully connected graphs than line graphs, given their highly connected nature with multiple, symmetric paths from one node to another. So, the question to ask then is what happens when order is imposed on a connected graph G:

$$\exists V' \subset V, \forall v \in V, E[\text{walkcost}(v)] \geq \frac{n}{2} + m$$

implies

$$\exists V'' \subset V, \forall v \in V, E[\text{walkcost}(v)] \leq \frac{n}{2} + m$$

As in the case of line graphs, forcing a subset of nodes to take on a cost greater than $n/2$ automatically forces an equal number of nodes to take on a cost lesser than $n/2$ in the same proportion, maintaining a global average of $n - 1/2$, this is owing to the symmetry and vertex transitivity of these graphs. Therefore, for any graph for which it can be argued in the following way:

- Label vertices of the $n \times n$ torus by $Z_n \times Z_n$. - Fix your two special nodes u and v. - Define

$$T: \ V \to V, \qquad T(x) \ = \ 2v - x \mod n,$$

i.e. "reflect x through v". Because addition is mod n, T is a graph-automorphism and $T^2 = \text{id}$.

For each $x \neq v$,

1. $h_x(u) = \Pr_u[\text{hit } x \text{ before } v]$. 2. By applying the automorphism T to a walk, one sees

$$\Pr_u[\text{hit } x \text{ before } v] \ = \ \Pr_v[\text{hit } T(x) \text{ before } u] =: h_{T(x)}(v).$$

3. But for *any* starting point y,

$$\Pr_y[\text{hit } x \text{ before } v] \;+\; \Pr_y[\text{hit } v \text{ before } x] \;=\; 1.$$

In particular, at $y = u$,

$$h_x(u) + \Pr_u[\text{hit } v \text{ before } x] = 1.$$

4. Yet by the same involution argument, $\Pr_u[\text{hit } v \text{ before } x] = h_{T(x)}(u)$.

$$h_x(u) + h_{T(x)}(u) = 1 \quad \text{for each pair } \{x, T(x)\}.$$

- There are n^2 total vertices; excluding v leaves $n^2 - 1$. - The involution T has no fixed points among $V \setminus \{v\}$ (since $x = T(x)$ would force $2x \equiv 2v$, i.e. $x \equiv v$). - Hence $V \setminus \{v\}$ splits into exactly

$$\frac{n^2 - 1}{2}$$

disjoint two-element orbits $\{x,\, T(x)\}$.

Summing $h_x(u) + h_{T(x)}(u) = 1$ over all these $\frac{n^2-1}{2}$ orbits gives

$$\sum_{x \neq v} h_x(u) \;=\; \frac{n^2 - 1}{2} \times 1 \;=\; \frac{n^2 - 1}{2}.$$

by circuit-walk correspondence,

$$\sum_{x \neq v} h_x(u) \;=\; E[\text{cost}_W(u, v)].$$

$\frac{n-1}{2}$ emerges as a central figure.

Conclusion and Future Directions:

These ideas now, conclude the matter of analysing the behaviour of probabilistic processess on graphs, as far as this thesis goes – whose scope has been limited to the behaviour of random

walks on regular graphs. Upon, establishing the basic setting, and notions of cost, our conjecture regarding randomized depth first search's cost was stated. As initial experiments had established, random walks provide a reasonable upperbound in the cost for randomized depth first search, and serves as a useful reference/pillar through which other probabilistic algorithms can be analyzed. Resistence network based characterizaiton of search cost put the phenomena of using random walks to explore graphs on a firm theoretical basis. Further more, a search for faster computation of cost for the toroidal grid has led, along with a faster computational method, to DFT diagonalizability, giving rise to the lattice green's funciton whose codification of distance effect seems to explain the asymptotic behaviour of random walks on regular graphs. Then, the symmetry and vertex transitivity properties of regular graphs have been leveraged to show why the cost hovers around $\frac{n-1}{2}$. In this light, the thesis project has taken on its promise to be an excercise in deconstructing the behaviour of random walk on regular graphs. However, as with all things exciting, there are more open questions than closed.

This work has taken upon the simple case of simple random walk on regular graphs. The central questions then are those of generalization as in what can be said about the behaviour of walks on say non transitive, but highly symmetric graphs? Can statements be made with respect to expected cost of a random graph? and so on. From an algorithmic perspective, whilst a relationship between rDFS and walk has been established, it is yet to be formalized; how does the cost change when the seeker/process is allowed to learn the strucutre of the graph as it explores is another exciting area to go off on. The extensive literature on loop erased random walks and statistical field theory is also not utilized yet. As such, there are a multitude of ways and manners in which the work presented here-in can be taken further.

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